



Bgp At Doors of Autonomous Systems is Simple (BADASS)

Summary: This document is a subject on network administration.

Version: 2.1

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Chapter I

Foreword



Chapter II

Introduction

The purpose of this project is to expand the knowledge you have gained through NetPractice. You will have to simulate a network and configure it using **GNS3 with docker images**.

BGP EVPN is based on BGP (RFC 4271) and its extensions, MP-BGP (RFC 4760). BGP is the routing protocol that drives the Internet. Through MP-BGP extensions, it can be used to carry reachability information (NLRI) for various protocols (IPv4, IPv6, L3 VPN and in this case, EVPN). EVPN is a special family used for publishing information about MAC addresses and the end devices that access them.

Chapter III

General guidelines

- The entire project must be completed in a **virtual machine**.
- This project involves installing and using **docker as well as GNS3**.
- You have to put all the configuration files of your project in folders located at the **root** of your repository (see section “Submission and peer-evaluation” for more information). The folders of the mandatory part will be named: **P1, P2 and P3**.



WARNING: This project involves many new concepts. Do not hesitate to take some time to read up on **how BGP and VXLANs work**. Many new terms are voluntarily used.



WARNING: This subject contains images that serve as examples and may contain typography that does not reflect what you should render.

Chapter IV

Mandatory part

This project involves setting up several environments under specific rules.

It is divided into three parts you have to do in the following order:

- Part 1: GNS3 configuration with Docker.
- Part 2: Discovering a VXLAN.
- Part 3: Discovering BGP with EVPN.



You must read the **entire subject** to understand what you need to do.

IV.1 Part 1: GNS3 configuration with Docker

For this first part, you must use and configure GNS3. It is thus necessary to **install and configure GNS3 as well as docker in your virtual machine**.

Now that everything works, you need to **use two docker images** which you must create.

The first image should be based on a system of your choice and must **contain at least busybox** or an equivalent solution.



Alpine seems to be a good solution.

The second image should use a system of your choice and must adhere to the following constraints:

- A software that manages packet routing (**zebra or quagga**).
- The **service BGPD active and configured**.
- The **service OSPFD active and configured**.
- An **IS-IS routing engine service**.
- **busybox** or an equivalent.



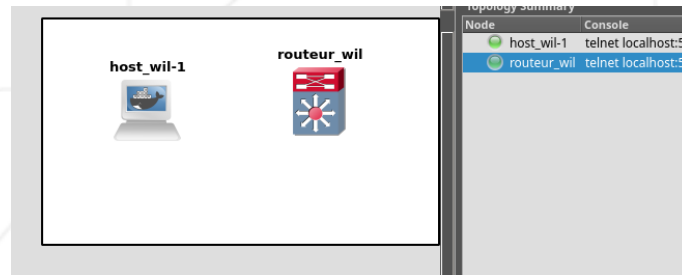
There are pre-built images which need to be configured with this kind of service. Your containers must work in GNS3 with the requested services. You can add anything you wish to complete this project.



Warning: Your images will be used throughout this project. No IP address should be configured by default.

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You must use these **two docker images in GNS3** and realize this small diagram. You need to have both machines working. We must be able to connect to them by GNS3.



The name of the machines is not put at random it will be necessary to have **your login in the name of each equipment** (here wil).

Below is an example of each image configured in GNS3:

```
/ # ps
PID  USER      TIME  COMMAND
    1  root         0:00  /bin/sh
    38  root         0:00  /gns3/bin/busybox script -qfc while true; do TERM=vt100 /g
    44  root         0:00  /bin/ash -c while true; do TERM=vt100 /gns3/bin/busybox sh
    45  root         0:00  /gns3/bin/busybox sh
    59  root         0:00  ps
/ # hostname
host_wil-1
/ #

/ # ps
PID  USER      TIME  COMMAND
    1  root         0:00  /sbin/tini -- /usr/lib/frr/docker-start
    38  root         0:00  /gns3/bin/busybox script -qfc while true; do TERM=vt100 /g
    45  root         0:00  /bin/ash -c while true; do TERM=vt100 /gns3/bin/busybox sh
    46  root         0:00  /gns3/bin/busybox sh
    94  root         0:00  tail -f /dev/null
   136  frr          0:00  /usr/lib/frr/zebra -d -F traditional -A 127.0.0.1 -s 90000
   141  frr          0:00  /usr/lib/frr/bgpd -d -F traditional -A 127.0.0.1
   148  frr          0:00  /usr/lib/frr/ospfd -d -F traditional -A 127.0.0.1
   151  frr          0:00  /usr/lib/frr/isisd -d -F traditional -A 127.0.0.1
   254  root         0:00  ps
/ # hostname
routeur_wil
/ #
```

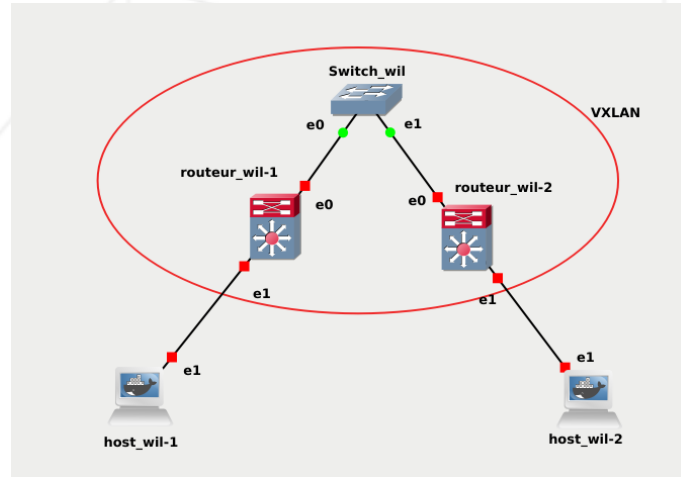
You must render this project in a **P1 folder** at the root of your git repository. You should also add the **configuration files with comments** to explain the set up of each equipment.



You must **export this project with a ZIP compression including the base images**. This file must be visible in your git repository

IV.2 Part 2: Discovering a VXLAN

You now have a functional basis to start setting up your first **VXLAN (RFC 7348) network**, **first in static mode** and **then in dynamic multicast mode**. Here is the topology of your first VXLAN:



You must **configure this network using a VXLAN with an ID of 10**, as shown in the examples below. You can use any VXLAN name you like here: **vxlan10**. You must set up a **bridge here: br0**. You can configure your **ETHERNET** interfaces as you wish. Below is an example of the expected result when we inspect the traffic between our two machines in our VXLAN.

```
# hostname
host_wil-1
# ifconfig eth1
eth1    Link encap:Ethernet  HWaddr DA:DA:AB:D7:54:C4
         inet addr:30.1.1.1  Bcast:0.0.0.0  Mask:255.255.255.0
         inet6 addr: fe80::d8da:abff:fed7:54c4/64 Scope:Link
         UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
         RX packets:50 errors:0 dropped:0 overruns:0 frame:0
         TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
         collisions:0 txqueuelen:1000
         RX bytes:3972 (3.8 KiB)  TX bytes:1216 (1.1 KiB)

# ping 30.1.1.2
PING 30.1.1.2 (30.1.1.2): 56 data bytes
64 bytes from 30.1.1.2: seq=0 ttl=64 time=0.443 ms
^C
--- 30.1.1.2 ping statistics ---
1 packets transmitted, 1 packets received, 0% packet loss
round-trip min/avg/max = 0.443/0.443/0.443 ms
#
```

```
# hostname
host_wil-2
# ifconfig eth1
eth1    Link encap:Ethernet  HWaddr 92:E6:7C:16:F3:18
         inet addr:30.1.1.2  Bcast:0.0.0.0  Mask:255.255.255.0
         inet6 addr: fe80::90e6:7cff:fe16:f318/64 Scope:Link
         UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
         RX packets:28 errors:0 dropped:0 overruns:0 frame:0
         TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
         collisions:0 txqueuelen:1000
         RX bytes:2180 (2.1 KiB)  TX bytes:1216 (1.1 KiB)
#
```

No.	Time	Source	Destination	Protocol	Length	Info
5	36.726751	30.1.1.1	30.1.1.2	ICMP	148	Echo (ping) request id=0x0039 seq=0/0
6	36.796971	30.1.1.2	30.1.1.1	ICMP	148	Echo (ping) reply id=0x0039 seq=0/0

```
> Frame 5: 148 bytes on wire (1184 bits), 148 bytes captured (1184 bits) on interface 0
> Ethernet II, Src: ca:f0:be:8d:3b:fa (ca:f0:be:8d:3b:fa), Dst: 46:41:50:cc:db:8f (46:41:50:cc:db:8f)
> Internet Protocol Version 4, Src: 10.1.1.1, Dst: 10.1.1.2
> User Datagram Protocol, Src Port: 59899, Dst Port: 4789
> Virtual extensible Local Area Network
  > Flags: 0x0800, VXLAN Network ID (VNI)
    Group Policy ID: 0
    VXLAN Network Identifier (VNI): 10
    Reserved: 0
  > Ethernet II, Src: da:da:ab:d7:54:c4 (da:da:ab:d7:54:c4), Dst: 92:e6:7c:16:f3:18 (92:e6:7c:16:f3:18)
> Internet Protocol Version 4, Src: 30.1.1.1, Dst: 30.1.1.2
> Internet Control Message Protocol
  0000  46 41 50 cc db 8f ca f6
  0010  00 06 50 a3 00 00 40 15
```

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We will now see the same thing using groups whose goal is to enable dynamic multicast.

The screenshot displays a network configuration and packet capture. At the top, a Wireshark window titled '*Standard input [router_wil-1 eth0 to Switch_wil Ethernet0]' shows a list of ICMPv6 packets. A red circle highlights a packet with source 30.1.1.1 and destination 30.1.1.2. Below the Wireshark window, two terminal windows are shown. The left terminal window shows the configuration of host_wil-1, including interface eth1 settings, IP address, and a ping command. The right terminal window shows the configuration of host_wil-2, including interface eth1 settings and IP address.

```
# ifconfig eth1
eth1      Link encap:Ethernet  HWaddr 12:5A:DE:F7:EA:53
          inet addr:30.1.1.1  Bcast:0.0.0.0  Mask:255.255.255.0
          inet6 addr: fe80::105a:deff:fef7:ea53/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:48 errors:0 dropped:0 overruns:0 frame:0
          TX packets:12 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3760 (3.6 KiB)  TX bytes:936 (936.0 B)

# # hostname
host_wil-1

# # ping 30.1.1.2
PING 30.1.1.2 (30.1.1.2): 56 data bytes
64 bytes from 30.1.1.2: seq=0 ttl=64 time=0.907 ms
64 bytes from 30.1.1.2: seq=1 ttl=64 time=0.573 ms
^C
--- 30.1.1.2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.573/0.740/0.907 ms
# #
```

```
# ifconfig eth1
eth1      Link encap:Ethernet  HWaddr EA:A6:14:12:8E:A4
          inet addr:30.1.1.2  Bcast:0.0.0.0  Mask:255.255.255.0
          inet6 addr: fe80::e8a6:14ff:fe12:8ea4/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:25 errors:0 dropped:1 overruns:0 frame:0
          TX packets:13 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1954 (1.9 KiB)  TX bytes:1006 (1006.0 B)

# # hostname
host_wil-2
# #
```

We can notice that our machines now have a group (here 239.1.1.1 you can modify this part):

```
# ip -d link show vxlan10
3: vxlan10: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue master br0 state UNKNOWN mode DEFAULT group default qlen 1000
    link/ether da:83:84:19:04:d5 brd ff:ff:ff:ff:ff:ff promiscuity 1 minmtu 68 maxmtu 65535
    vxlan id 10 group 239.1.1.1 dev eth0 srcport 0 dstport 4789 ttl auto ageing 300 udpchecksum noudp6zerocsumtx noudp6zerocsumrx
    bridge_slave state forwarding priority 32 cost 100 hairpin off guard off root_block off fastleave off learning on flood on flood on
    port_id 0x8002 port_no 0x2 designated_port 32770 designated_cost 0 designated_bridge 8000.DA:83:84:19:04:D5 designated_root 8000.DA:83:84:19:04:D5
    hold timer 0.00 message_age_timer 0.00 forward_delay_timer 0.00 topology_change_ack 0 config_pending 0 proxy_arp off proxy_arp_wifi off mcast_router 1 mcast_fast_leave off mcast_flood on mcast_to_unicast off neigh_suppress off group_fwd_mask 0 group_fwd_mask_str 0x0 vlan_tunnel off isolated off addrgenmode eui64 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535
# # hostname
router_wil-1
# #
```

```
# ip -d link show vxlan10
3: vxlan10: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue master br0 state UNKNOWN mode DEFAULT group default qlen 1000
    link/ether ae:df:08:4c:0c:0d brd ff:ff:ff:ff:ff:ff promiscuity 1 minmtu 68 maxmtu 65535
    vxlan id 10 group 239.1.1.1 dev eth0 srcport 0 dstport 4789 ttl auto ageing 300 udpchecksum noudp6zerocsumtx noudp6zerocsumrx
    bridge_slave state forwarding priority 32 cost 100 hairpin off guard off root_block off fastleave off learning on flood on flood on
    port_id 0x8002 port_no 0x2 designated_port 32770 designated_cost 0 designated_bridge 8000.02:AF:63:0B:5E:00 designated_root 8000.02:AF:63:0B:5E:00
    hold timer 0.00 message_age_timer 0.00 forward_delay_timer 0.00 topology_change_ack 0 config_pending 0 proxy_arp off proxy_arp_wifi off mcast_router 1 mcast_fast_leave off mcast_flood on mcast_to_unicast off neigh_suppress off group_fwd_mask 0 group_fwd_mask_str 0x0 vlan_tunnel off isolated off addrgenmode eui64 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535
# # hostname
router_wil-2
# #
```

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Below is an example of how to display our mac address table in our two routers:

```

/usr/bin/zsh 59x24
# hostname
routeur_wil-1
# brctl showmacs br0
port no mac addr      is local?   ageing time
#
2  02:af:63:db:5e:00    no          136.98
1  12:5a:de:f7:ea:53    no          202.52
2  ae:df:08:4c:0c:0d    no          196.37
2  da:83:84:19:04:d5    yes         0.00
2  da:83:84:19:04:d5    yes         0.00
2  ea:a6:14:12:8e:a4    no          215.32
1  ee:3f:ab:b3:36:38    yes         0.00
1  ee:3f:ab:b3:36:38    yes         0.00
#

/usr/bin/zsh 63x24
# hostname
routeur_wil-2
# brctl showmacs br0
port no mac addr      is local?   ageing timer
#
1  02:af:63:db:5e:00    yes         0.00
1  02:af:63:db:5e:00    yes         0.00
2  12:5a:de:f7:ea:53    no          169.23
2  ae:df:08:4c:0c:0d    yes         0.00
2  ae:df:08:4c:0c:0d    yes         0.00
2  da:83:84:19:04:d5    no          21.77
1  ea:a6:14:12:8e:a4    no          182.03
#
```

You must render this project in a P2 folder at the root of your git repository. You should also add the configuration files with comments to explain the set up of each equipment.



You must export this project with a ZIP compression including the base images. This file must be visible in your git repository.



You must use correct and consistent names for your equipment here with the login of one of the group members.

IV.3 Part 3: Discovering BGP with EVPN

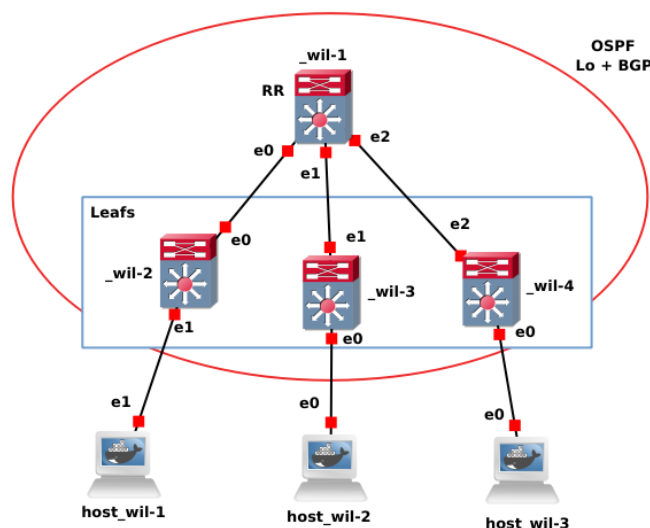
Now that you have mastered the basic principles of VXLAN, we will go a little further and explore the principles of the BGP EVPN (RFC 7432) without using MPLS to simplify the process. The controller will learn the MAC addresses automatically. We will use our VXLAN with ID 10, as seen in the previous part.

As in the second part we start with the topology of the expected network. We will use the principle of route reflection (=RR). Our leaves (VTEP) will be configured to have dynamic relationships.

This diagram represents a small data center.



For the sake of readability the names are shorter here. You will have to use OSPF to simplify the evaluation.



We can see our visibility from our VTEP **_wil-4** the 3 VTEPs 1.1.[1.4]

```
wil-4(config-router)# do sh ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

O>* 1.1.1.1/32 [110/10000] via 10.1.1.9, eth2, weight 1, 00:00:44
O>* 1.1.1.3/32 [110/20000] via 10.1.1.9, eth2, weight 1, 00:00:44
O  1.1.1.4/32 [110/0] is directly connected, lo, weight 1, 00:00:58
C>* 1.1.1.4/32 is directly connected, lo, 00:00:58
O>* 10.1.1.0/30 [110/20000] via 10.1.1.9, eth2, weight 1, 00:00:44
O>* 10.1.1.4/30 [110/20000] via 10.1.1.9, eth2, weight 1, 00:00:44
O  10.1.1.8/30 [110/10000] is directly connected, eth2, weight 1, 00:00:49
C>* 10.1.1.8/30 is directly connected, eth2, 00:00:58
wil-4(config-router)#
```

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We have only one route for the moment with our controller (RR):

```
wil-4(config-router)# do sh bgp summary

IPv4 Unicast Summary:
BGP router identifier 1.1.1.4, local AS number 1 vrf-id 0
BGP table version 0
RIB entries 0, using 0 bytes of memory
Peers 1, using 14 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  PfxSnt
1.1.1.1        4        1         7         7         0     0     0 00:02:12      0         0

Total number of neighbors 1

L2VPN EVPN Summary:
BGP router identifier 1.1.1.4, local AS number 1 vrf-id 0
BGP table version 0
RIB entries 1, using 192 bytes of memory
Peers 1, using 14 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  PfxSnt
1.1.1.1        4        1         7         7         0     0     0 00:02:12      0         1

Total number of neighbors 1
wil-4(config-router)#
```

When there is no host running we can see our VNI (10 here) as well as our pre-configured routes (type 3). No route type 2 seems to exist and it is quite normal.

```
wil-4(config-router)# do sh bgp l2vpn evpn
BGP table version is 1, local router ID is 1.1.1.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN type-1 prefix: [1]:[ESI]:[EthTag]:[IPlen]:[VTEP-IP]
EVPN type-2 prefix: [2]:[EthTag]:[MAClen]:[MAC]:[IPlen]:[IP]
EVPN type-3 prefix: [3]:[EthTag]:[IPlen]:[OrigIP]
EVPN type-4 prefix: [4]:[ESI]:[IPlen]:[OrigIP]
EVPN type-5 prefix: [5]:[EthTag]:[IPlen]:[IP]

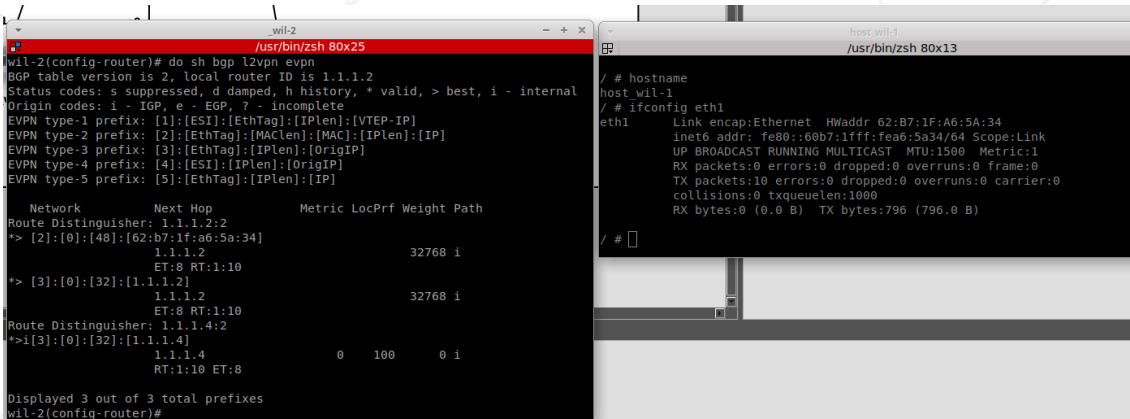
  Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 1.1.1.2:2
*>i[3]:[0]:[32]:[1.1.1.2]
  1.1.1.2              0      100      0 i
  RT:1:10 ET:8
Route Distinguisher: 1.1.1.4:2
*> [3]:[0]:[32]:[1.1.1.4]
  1.1.1.4              32768 i
  ET:8 RT:1:10

Displayed 2 out of 2 total prefixes
wil-4(config-router)#
```

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A machine `host_wil-1` is now functional. We can notice that without assigning an IP address our VTEP (`wil_2`) automatically discovers the MAC address of the functional machines.

We can also see the automatic creation of a route type 2:



```
wil-2(config-router)# do sh bgp l2vpn evpn
BGP table version is 2, local router ID is 1.1.1.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN type-1 prefix: [1]:[ESI]:[EthTag]:[IPlen]:[VTEP-IP]
EVPN type-2 prefix: [2]:[EthTag]:[MAClen]:[MAC]:[IPlen]:[IP]
EVPN type-3 prefix: [3]:[EthTag]:[IPlen]:[OrigIP]
EVPN type-4 prefix: [4]:[ESI]:[IPlen]:[OrigIP]
EVPN type-5 prefix: [5]:[EthTag]:[IPlen]:[IP]

  Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 1.1.1.2:2
*> [2]:[0]:[48]:[62:b7:1f:a6:5a:34]
      1.1.1.2
      ET:8 RT:1:10
      32768 i
*> [3]:[0]:[32]:[1.1.1.2]
      1.1.1.2
      ET:8 RT:1:10
      32768 i
Route Distinguisher: 1.1.1.4:2
*> [3]:[0]:[32]:[1.1.1.4]
      1.1.1.4
      RT:1:10 ET:8
      0 100 0 i
Displayed 3 out of 3 total prefixes
wil-2(config-router)#
```

```
host_wil-1
/usr/bin/zsh 80x13

/ # hostname
host_wil-1
/ # ifconfig eth1
eth1      Link encap:Ethernet  HWaddr 62:B7:1F:A6:5A:34
          inet6 addr: fe80::60b7:1fff:fe6a:5a34/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:10 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:796 (796.0 B)

/ #
```

In the same way, when we look at a second VTEP(`_wil-4`), we can notice the creation of a new route type 2 generated by our RR:



```
wil-4(config-router)# do sh bgp l2vpn evpn
BGP table version is 2, local router ID is 1.1.1.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN type-1 prefix: [1]:[ESI]:[EthTag]:[IPlen]:[VTEP-IP]
EVPN type-2 prefix: [2]:[EthTag]:[MAClen]:[MAC]:[IPlen]:[IP]
EVPN type-3 prefix: [3]:[EthTag]:[IPlen]:[OrigIP]
EVPN type-4 prefix: [4]:[ESI]:[IPlen]:[OrigIP]
EVPN type-5 prefix: [5]:[EthTag]:[IPlen]:[IP]

  Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 1.1.1.2:2
*> i[2]:[0]:[48]:[62:b7:1f:a6:5a:34]
      1.1.1.2
      0 100 0 i
      RT:1:10 ET:8
*> i[3]:[0]:[32]:[1.1.1.2]
      1.1.1.2
      0 100 0 i
      RT:1:10 ET:8
Route Distinguisher: 1.1.1.4:2
*> [3]:[0]:[32]:[1.1.1.4]
      1.1.1.4
      ET:8 RT:1:10
      32768 i
Displayed 3 out of 3 total prefixes
_wil-4(config-router)#
```

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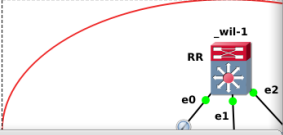
We repeat the operation with a second machine (host_wil-3). We can notice the second route set up by type 2. There is no assignment of IP address:

```
wil-2(config-router)# do sh bgp l2vpn evpn
BGP table version is 2, local router ID is 1.1.1.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN type-1 prefix: [1]:[ESI]:[EthTag]:[IPlen]:[VTEP-IP]
EVPN type-2 prefix: [2]:[EthTag]:[MAClen]:[MAC]:[IPlen]:[IP]
EVPN type-3 prefix: [3]:[EthTag]:[IPlen]:[OrigIP]
EVPN type-4 prefix: [4]:[ESI]:[IPlen]:[OrigIP]
EVPN type-5 prefix: [5]:[EthTag]:[IPlen]:[IP]

  Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 1.1.1.2:2
*> [2]:[0]:[48]:[62:b7:1f:a6:5a:34]
      1.1.1.2                      32768 i
      ET:8 RT:1:10
*> [3]:[0]:[32]:[1.1.1.2]
      1.1.1.2                      32768 i
      ET:8 RT:1:10
Route Distinguisher: 1.1.1.4:2
*>i[2]:[0]:[48]:[1e:80:95:23:b8:45]
      1.1.1.4                      0    100    0 i
      RT:1:10 ET:8
*>i[3]:[0]:[32]:[1.1.1.4]
      1.1.1.4                      0    100    0 i
      RT:1:10 ET:8

Displayed 4 out of 4 total prefixes
wil-2(config-router)#
```

For our verification a simple ping allows us to see that we can access all the machines through our RR using the VTEPs. We can see the VXLAN configured to 10 as well as our packets ICMP. We also see packets OSPF configured:



```
host_wil-1 /usr/bin/zsh 80x22
# hostname
host_wil-1
# ping 20.1.1.2
PING 20.1.1.2 (20.1.1.2): 56 data bytes
64 bytes from 20.1.1.2: seq=0 ttl=64 time=0.993 ms
64 bytes from 20.1.1.2: seq=1 ttl=64 time=0.660 ms
64 bytes from 20.1.1.2: seq=2 ttl=64 time=0.612 ms
^C
--- 20.1.1.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.612/0.755/0.993 ms
# ifconfig eth1
eth1  Link encap:Ethernet  HWaddr 62:B7:1F:A6:5A:34
       inet addr:20.1.1.1 Bcast:0.0.0.0 Mask:255.255.255.0
       inet6 addr: fe80::60b7:1fff:fe6d:5a34/64 Scope:Link
       UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
       RX packets:22 errors:0 dropped:0 overruns:0 frame:0
       TX packets:18 errors:0 dropped:0 overruns:0 carrier:0
       collisions:0 txqueueLen:1000
       RX bytes:1664 (1.6 KiB)  TX bytes:1384 (1.3 KiB)

#
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ea:1e:d4:81:84:a6	2a:a6:79:c3:92:a0	ARP	42	Who has 10.1.1.2? Tell 10.1.1.1
2	0.000079	2a:a6:79:c3:92:a0	ea:1e:d4:81:84:a6	ARP	42	10.1.1.2 is at 2a:a6:79:c3:92:a0
3	4.608091	10.1.1.2	224.0.0.5	OSPF	82	Hello Packet
4	5.067560	10.1.1.1	224.0.0.5	OSPF	82	Hello Packet
5	14.440283	62:b7:1f:a6:5a:34	Broadcast	ARP	92	Who has 20.1.1.2? Tell 20.1.1.1
6	14.449575	1e:80:95:23:b8:45	62:b7:1f:a6:5a:34	ARP	92	20.1.1.2 is at 1e:80:95:23:b8:45
7	14.449728	20.1.1.1	20.1.1.2	ICMP	148	Echo (ping) request id=0x0029, seq=0
8	14.450029	20.1.1.2	20.1.1.1	ICMP	148	Echo (ping) reply id=0x0029, seq=0
9	14.611476	10.1.1.2	224.0.0.5	OSPF	82	Hello Packet
10	15.067560	10.1.1.1	224.0.0.5	OSPF	82	Hello Packet
11	15.450339	20.1.1.1	20.1.1.2	ICMP	148	Echo (ping) request id=0x0029, seq=1
12	15.450741	20.1.1.2	20.1.1.1	ICMP	148	Echo (ping) reply id=0x0029, seq=1
13	16.455903	20.1.1.1	20.1.1.2	ICMP	148	Echo (ping) request id=0x0029, seq=2
14	16.456291	20.1.1.2	20.1.1.1	ICMP	148	Echo (ping) reply id=0x0029, seq=2
15	19.456560	1e:80:95:23:b8:45	62:b7:1f:a6:5a:34	ARP	92	Who has 20.1.1.1? Tell 20.1.1.2
16	19.456742	62:b7:1f:a6:5a:34	1e:80:95:23:b8:45	ARP	92	20.1.1.1 is at 62:b7:1f:a6:5a:34
17	24.580155	ea:1e:d4:81:84:a6	2a:a6:79:c3:92:a0	ARP	42	Who has 10.1.1.2? Tell 10.1.1.1
18	24.580223	2a:a6:79:c3:92:a0	ea:1e:d4:81:84:a6	ARP	42	10.1.1.2 is at 2a:a6:79:c3:92:a0
19	24.611523	10.1.1.2	224.0.0.5	OSPF	82	Hello Packet
20	25.068015	10.1.1.1	224.0.0.5	OSPF	82	Hello Packet

```
Frame 7: 148 bytes on wire (1184 bits), 148 bytes captured (1184 bits) on interface 0
  Ethernet II, Src: 2a:a6:79:c3:92:a0 (2a:a6:79:c3:92:a0), Dst: ea:1e:d4:81:84:a6 (ea:1e:d4:81:84:a6)
  Internet Protocol Version 4, Src: 10.1.1.2, Dst: 10.1.1.10
  User Datagram Protocol, Src Port: 53458, Dst Port: 4789
  Virtual extensible Local Area Network
    Flags: 0x0000, VLAN Network ID (VNI)
    Group Policy ID: 0
  VXLAN Network Identifier (VNI): 10
  Reserved: 0
  Ethernet II, Src: 62:b7:1f:a6:5a:34 (62:b7:1f:a6:5a:34), Dst: 1e:80:95:23:b8:45 (1e:80:95:23:b8:45)
  Internet Protocol Version 4, Src: 20.1.1.1, Dst: 20.1.1.2
  Internet Control Message Protocol
```

```
host_wil-3 /usr/bin/zsh 80x12
# ifconfig eth0
eth0  Link encap:Ethernet  HWaddr 1E:80:95:23:B8:45
       inet addr:20.1.1.2 Bcast:0.0.0.0 Mask:255.255.255.0
       inet6 addr: fe80::1c80:95ff:fe23:b845/64 Scope:Link
       UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
       RX packets:8 errors:0 dropped:0 overruns:0 frame:0
       TX packets:17 errors:0 dropped:0 overruns:0 carrier:0
       collisions:0 txqueueLen:1000
       RX bytes:588 (588.0 B)  TX bytes:1314 (1.2 KiB)
```

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You must render this project in a P3 folder at the root of your git repository. You should also add the configuration files with comments to explain the set up of each equipment.



You must export this project with a ZIP compression including the image bases. This file must be visible in your git repository.



You must use correct and consistent names for your equipment here with the login of one of the group members.

Chapter V

Submission and peer-evaluation

Submit your assignment to your `Git` repository in the usual manner. Only the work inside your repository will be evaluated during the defense. Do not hesitate to double-check the names of your folders and files to ensure they are correct.

Reminder:

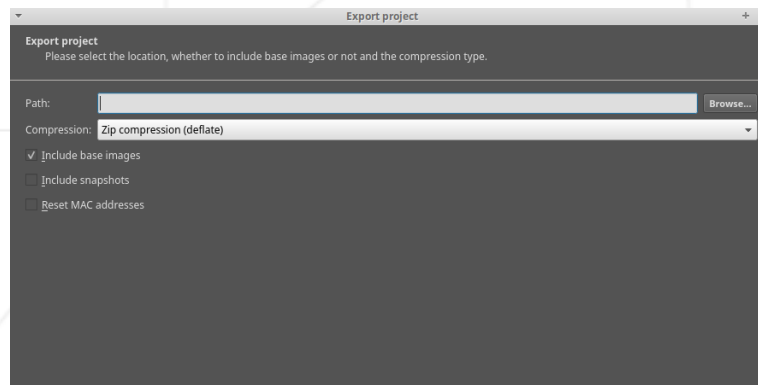
- Turn the mandatory part in three folders located at the root of your repository: P1, P2 et P3.

Below is an example of the expected directory structure:

```
> find -maxdepth 2 -ls
424242      4 drwxr-xr-x  6 wandre  wil42      4096 sept. 17 23:42 .
424242      4 drwxr-xr-x  3 wandre  wil42      4096 sept. 17 23:42 ./P1
424242      4 -rw-r--r--  1 wandre  wil42      XXXX sept. 17 23:42 ./P1/P1.gns3project
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P1/_wil-1_host
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P1/_wil-2
424242      4 drwxr-xr-x  3 wandre  wil42      4096 sept. 17 23:42 ./P2
424242      4 -rw-r--r--  1 wandre  wil42      XXXX sept. 17 23:42 ./P2/P2.gns3project
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P2/_wil-1_g
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P2/_wil-1_host
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P2/_wil-1_s
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P2/_wil-2_g
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P2/_wil-2_host
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P2/_wil-2_s
424242      4 drwxr-xr-x  3 wandre  wil42      4096 sept. 17 23:42 ./P3
424242      4 -rw-r--r--  2 wandre  wil42      4096 sept. 17 23:42 ./P3/P3.gns3project
424242      4 -rw-r--r--  2 wandre  wil42      4096 sept. 17 23:42 ./P3/_wil-1
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P3/_wil-1_host
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P3/_wil-2
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P3/_wil-2_host
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P3/_wil-3
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P3/_wil-3_host
424242      4 -rw-r--r--  2 wandre  wil42      XXXX sept. 17 23:42 ./P3/_wil-4
> file P3/P3.gns3project
P3/P3.gns3project: Zip archive data, at least v2.0 to extract
```

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To export your projects in zip format go to the menu file then export portable project:



During the evaluation you may have to explain the terms used in the subject. We strongly encourage you to take the time to understand each of these.



The evaluation process will happen on the computer of the evaluated group.